

Pillow `PcfFontFile.\_load\_bitmaps()`: `Image.frombytes()` called without `\_decompression\_bomb\_check()` — bomb protection bypass via PCF font loading

Report Type: Weekly · Generated: July 20, 2026 21:27 UTC · Coverage: Jul 20 - Jul 20, 2026

1 Total Advisories	0 Critical	1 High	0 Medium
HIGH Severity	7.5 CVSS / Risk Score	TLP:CLEAR TLP Classification	PRO Customer Tier

Executive Summary

Description `PIL/PcfFontFile.py` `\_load\_bitmaps()` (line 227) reads glyph dimensions from the PCF `METRICS` section and passes them directly to `Image.frombytes()` without calling `Image.\_decompression\_bomb\_check()`. Dimensions originate from unsigned 16-bit values: `` xsize = right - left (max: 65535 - 0 = 65535) ysize = ascent + descent (max: 65535 + 65535 = 131070) `` Maximum exploitable pixel count: ****65,535 × 131,070 = 8,589,734,450 pixels**** — ****48× the DecompressionBombError threshold****. ****Vulnerable code**** (`PIL/PcfFontFile.py` line 224–227):****** ``python for i in range(nbitmaps): xsize, ysize = metrics[i]:2] # from PCF METRICS — attacker-controlled b, e = offsets[i : i + 2] bitmaps.append( Image.frombytes("1", (xsize, ysize), data[b:e], "raw", mode, pad(xsize)) # ↑ NO \_decompression\_bomb\_check()! ) `` `Image.frombytes()` calls `Image.new()` first (allocating the full C-heap buffer), ****then**** attempts to fill it. This crea

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Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
Pillow `PcfFontFile._load_bitmaps()`: `Image.frombytes` called with 0.5	Critical	10.5	35.0%	## Description `PIL/PcfFontFile.py` `_load_bitmaps()` (line 227) reads glyph dimensions from a file.

MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Count	Associated CVEs / Indicators
CDB-UNATTR-CVE	1	-

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intel/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - CDB-ADVISORY Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: Detects exploitation indicators related to CDB-ADVISORY.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/07/20
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CDB-ADVISORY'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - Pillow `PcfFontFile._load_bitmaps()`: `Image.frombytes()` called without
`_decompression_bomb_check()` - bomb protection bypass via PCF font loading
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "Pillow `PcfFontFile._load_bitmaps()`: `I"
or AlertName has "Pillow `PcfFontFile._load_bitmaps()`: `I"
or ExtendedProperties has "Pillow `PcfFontFile._load_bitmaps()`: `I"
| extend Rule = "APEX-ADVISORY", Severity = "HIGH"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield Pillow `PcfFontFile.\_load\_bitmaps()`: `Image.frombytes()` called without `\_decompression\_bomb\_check()` — bomb protection bypass via PCF font loading instances exposed to the internet.

- 3. Enable verbose access logging on all Pillow `PcfFontFile._load_bitmaps()`: `Image.frombytes()` called without `_decompression_bomb_check()` — bomb protection bypass via PCF font loading endpoints immediately.
- 4. Pull last 72h of web server and application logs for forensic baselining.
- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for Pillow `PcfFontFile._load_bitmaps()`: `Image.frombytes()` called without `_decompression_bomb_check()` — bomb protection bypass via PCF font loading or implement recommended mitigation.
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

\$800K - \$4.5M Breach Cost Range (FAIR)	\$1.8M IBM Cost of Breach 2025 Median	\$450K/day Business Interruption Cost	\$320K Regulatory Fine Exposure
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High severity incidents activate IR retainers, forensic investigation (avg 47 days), and reputational damage impacting 12-18% of revenue.

Remediation Cost Estimate: \$180K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
GDPR	Art. 32 / Art. 33	Appropriate technical measures; notify DPA if personal data affected

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Framework	Reference	Obligation
PCI-DSS v4.0	Req. 6.3.3	Patches within 1 month; documented risk acceptance if deferred
NIS2 Directive	Art. 21	Risk management measures including vulnerability handling policies
NIST CSF 2.0	RS.MI / RC.RP	Incident mitigation and recovery planning; post-incident review

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Apply patches immediately for Pillow ``PcfFontFile._load_bitmaps()`: `Image.frombytes()`` called without `_decompression_bomb_check()`` — bomb protection bypass via PCF font loading.`
2. Enable enhanced monitoring for related IOCs.
3. Review SIEM rules for associated MITRE ATT&CK techniques.
4. Apply vendor patch for Pillow ``PcfFontFile._load_bitmaps()`: `Image.frombytes()`` called without `_decompression_bomb_check()`` — bomb protection bypass via PCF font loading immediately - check NVD for official fix guidance.`
5. Enable enhanced logging and alerting on all systems running affected software versions.
6. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
7. Audit all internet-facing instances of affected products and confirm patch deployment.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--4b420bd12b8920d193eab864
Report Type	Weekly
Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
Generated At	2026-07-20T21:27:04.06330
Customer Tier	PRO
Customer Email	-
TLP	TLP: CLEAR
STIX Version	2.1
ATT&CK Version	v15

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